THE BEGINNER'S GUIDE TO

# Caravan Solar

Clear recommendations, brand comparisons, and shopping lists for Australian caravanners who don't want to become electrical engineers.

WHAT'S INSIDE

- Your power audit in 10 minutes
- Exactly what to buy, component by component
- Brand comparisons (REDARC, Victron, Enerdrive, Renogy)
- Ready-to-use shopping lists for your tier
- Printable checklists and worksheets

**Sized for how you travel:**

Weekender | Tourer | Full-timer

WHAT'S INSIDE

# Contents



LOOK FOR THESE BOXES

SKIP TO THIS

Every section has a callout box like this one. Read only these boxes and you'll have your answer in about 5 minutes.

# The 60-Second Version

SKIP THIS IF YOU WANT

SKIP TO THIS

Just want the "what to buy" part? Go straight to Section 3. Otherwise, here's the 60-second version.

## The Whole System in Plain English

1. Solar panels sit on your roof and turn sunlight into power.
2. A battery stores that power. This is the fuel tank.
3. A charge controller manages charging safely.
4. A DC-DC charger tops up the battery from your car while you drive.
5. An inverter lets you plug in normal 240V things (laptop, microwave).
6. A battery monitor is your fuel gauge.

**That's it. Six things. The rest of this guide tells you exactly which ones to buy and what size.**

## One Quick Concept

Watt-hours measure daily power use. If something uses 60 watts for 10 hours, that's 600 Wh. Your fridge uses about 60W all day: roughly 1,440 Wh.

That daily number drives everything: battery size, solar, the lot. Section 2 helps you work out yours in about 10 minutes.

SECTION 2

# Your Power Audit

We need to figure out roughly how much power your caravan uses in a day. Takes about 10 minutes, and it's the one piece of homework in this guide. Everything else flows from it.

Don't stress about being exact. Within 20% is plenty. We add a buffer for the stuff you forget.

## Walk-Through: Dave and Sarah's Setup

Dave and Sarah are tourers. Weeks on the road, mix of parks and free camping. Here's their typical day off-grid:

| Appliance | Usage | Daily Wh |
|---|---|---|
| Fridge (all day, all night) | 60W x 24h | ~1,440 Wh |
| LED lights (evening) | 30W x 5h | ~150 Wh |
| TV after dinner | 40W x 3h | ~120 Wh |
| Phones + laptop | 100W x 2h | ~220 Wh |
| Water pump, heater, etc. | Various | ~220 Wh |

**Daily total: about 2,150 Wh. Add 25% buffer: about 2,700 Wh per day.**

That's it. No complicated formulas. Just: what do you have, how long is it on, roughly how much does it use?

Your turn. Use the Power Audit Worksheet at the back. It lists every common caravan appliance with typical power draws.

## Which Tier Are You?

Your daily number tells you what kind of system you need:

**WEEKENDER** — $1,500 - $2,500

### Under 1,500 Wh/day

Short trips, mostly caravan parks, some free camping. Fridge, lights, phones.

**TOURER** — $3,000 - $5,000

### 1,500 to 3,000 Wh/day

Weeks on the road, mix of parks and free camping. Fridge, lights, TV, laptops.

**FULL-TIMER** — $5,000 - $8,000+

### Over 3,000 Wh/day

Living on the road. Everything above plus microwave, coffee machine, hair dryer.

**A NOTE ON AIR CONDITIONING** — SKIP TO THIS

Air con off-grid isn't practical on a standard setup. If it's a must-have, you need a 48V system ($10,000+) or a generator. This guide covers standard 12V setups.

**Got your number? Good. From here on, the guide is built around your tier. Every recommendation, every shopping list is matched to Weekender, Tourer, or Full-timer.**

# What to Buy

Each component gets a quick answer at the top. Trust us? Read the callout boxes and move on.

## Battery

BOTTOM LINE

SKIP TO THIS

Buy lithium. Not AGM, not gel. Lithium. Costs more upfront but lasts years longer and gives way more usable power. 200Ah lithium is the sweet spot for most people.

An AGM "200Ah" battery only gives about 100Ah usable. A lithium "200Ah" gives 160-200Ah usable. You'd need two AGMs to match one lithium, and they weigh twice as much and last 3-5 years vs 10+.

**The most common beginner mistake is buying AGM now and upgrading to lithium a year later. That's $400-$800 in the bin.**

### What to buy by tier:

- Weekender: 100-200Ah lithium
- Tourer: 200Ah lithium (one battery does the job)
- Full-timer: 300-400Ah lithium

Make sure it has a built-in BMS (Battery Management System). All the brands we recommend include one.

## Solar Panels

**BOTTOM LINE** SKIP TO THIS

Fixed panels on the roof PLUS a portable folding panel. Roof panels do daily work; portable is for shade. Go monocrystalline.

Fixed panels bolt to your roof and work all day. Portable panels fold out at camp and angle toward the sun. Most people get both.

A "400W" panel produces 200-280W in real conditions. This guide's sizing accounts for that.

**By tier:**

- Weekender: 200W fixed
- Tourer: 300-400W fixed + 160-200W portable
- Full-timer: 400-600W fixed + 200W+ portable

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## Charge Controller

**BOTTOM LINE** SKIP TO THIS

Under 200W solar: PWM is fine. Everyone else: get MPPT. It squeezes 15-30% more power. Size at about 1/3 of your solar wattage in amps.

**By tier:**

- Weekender: 20A PWM or MPPT
- Tourer: 30-40A MPPT
- Full-timer: 40-50A MPPT

## DC-DC Charger

**BOTTOM LINE** SKIP TO THIS

Charges your battery from the car while you drive. Not optional. A few hours of driving puts a big chunk back in. Buy a REDARC BCDC.

Modern cars have smart alternators that don't work with a direct connection. A DC-DC charger sorts this out. A 40A unit gives 120-160Ah on a 3-4 hour drive.

**By tier:**

- Weekender: 20A DC-DC charger
- Tourer: 40A DC-DC charger
- Full-timer: 40-60A DC-DC charger

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## Inverter

**BOTTOM LINE** SKIP TO THIS

Pure sine wave inverter. NEVER modified sine wave. Size for your biggest appliance. Weekender: 1000W. Tourer: 2000W. Full-timer: 3000W.

Your battery is 12V, appliances need 240V. The inverter converts it. Always pure sine wave. Modified sine wave saves $50-$100 but damages laptops and chargers.

How big? Biggest single appliance: phones and laptops = 1000W. Microwave = 2000W. Hair dryer = 3000W.

## Battery Monitor

**BOTTOM LINE** SKIP TO THIS

Buy a Victron SmartShunt ($130-$160). Shows battery level on your phone via Bluetooth. Without a monitor, you're driving without a fuel gauge.

It shows exactly what percentage you're at, how fast you're using power, and how long until full. Without it you'll wear out the battery without realising.

## Your Complete System at a Glance

All six components. Find your tier:

| Component | Weekender | Tourer | Full-timer |
|---|---|---|---|
| **Battery** | 100-200Ah | 200Ah | 300-400Ah |
| **Solar (roof)** | 200W | 300-400W | 400-600W |
| **Solar (portable)** | Optional | 160-200W | 200W+ |
| **Controller** | 20A PWM/MPPT | 30-40A MPPT | 40-50A MPPT |
| **DC-DC charger** | 20A | 40A | 40-60A |
| **Inverter** | 1000W PSW | 2000W PSW | 3000W PSW |
| **Monitor** | SmartShunt | SmartShunt | SmartShunt |

**That's your whole setup. Six things, sized for how you travel. Next section turns this into a shopping list with specific brands and prices.**

# Which Brands to Buy

**BOTTOM LINE** SKIP TO THIS

REDARC for DC-DC chargers. Victron for monitoring. Enerdrive for value. Renogy if budget is tight. Mixing brands is fine.

**This guide isn't sponsored by or affiliated with any brand. Genuine recommendations based on what holds up.**

## REDARC

The Safe Choice

Australian-made, based in Adelaide. Premium price, excellent build quality. Their BCDC DC-DC charger range is the industry standard.

Buy REDARC for: DC-DC chargers. Think twice for: monitoring (Victron's app is better), and if budget is tight.

## Victron

The Smart Choice

Dutch company, since 1975. Best app in the business. Battery level on your phone, graphs, alerts. Victron's ecosystem is in a league of its own.

Buy Victron for: charge controllers, battery monitors, inverters. Think twice for: more separate components than REDARC or Enerdrive.

## Enerdrive

The Value Choice

Australian brand (now Dometic). Good quality at lower price than REDARC, 5-year warranty, lifetime tech support. Their eSystem pre-wired boards are close to plug-and-play.

Buy for: batteries, inverters, pre-wired systems. Think twice: narrower product range.

## Renogy

The Budget Choice

Chinese brand with AU warehouse. Significantly cheaper. A 200Ah lithium for under $500 vs $800+ elsewhere. Good for Weekenders on a tight budget.

Buy for: panels, batteries, starter kits. Think twice: full-timer daily reliance, slower warranty support.

# One-Page Comparison

| | REDARC | Victron | Enerdrive | Renogy |
|---|---|---|---|---|
| Made in | Australia | Netherlands | AU (Dometic) | China |
| Price | Premium | Premium | Mid-range | Budget |
| Best at | DC-DC | App + monitor | Value + install | Price |
| Warranty | 2 years | 5 years | 5yr + lifetime | 5 years |
| App | Good | Excellent | Good | Basic |
| AU support | Excellent | Good | Excellent | OK |

## Mixing Brands

You don't have to buy from one brand. Mix and match: REDARC DC-DC, Victron monitoring, Enerdrive battery. Standard connections, not proprietary.

## Where to Buy in Australia

- Solar 4 RVs, Caravan RV Camping, My Generator (specialist retailers)
- REDARC, Enerdrive, Renogy sell direct. Victron via distributors.
- Jaycar for cables, fuses, and small bits.

**Avoid no-name brands on eBay/Amazon with no Australian warranty.**

DON'T FORGET THE SMALL BITS

SKIP TO THIS

Everyone forgets cables, fuses, Anderson plugs, and mounting brackets. Budget $100-$400. The Shopping List Worksheet has a full checklist.

WEEKENDER SHOPPING LIST $1,500 - $2,500

| What | Buy This | Cost |
|---|---|---|
| Battery | Renogy/Enerdrive 200Ah Li | $480-$900 |
| Solar (roof) | 1x 200W monocrystalline | $200-$350 |
| Controller | 20A MPPT (Renogy/Victron) | $100-$200 |
| DC-DC charger | REDARC BCDC1220 (20A) | $200-$400 |
| Inverter | 1000W pure sine wave | $150-$300 |
| Monitor | Victron SmartShunt | $130-$160 |
| Wiring + bits | Cable, fuses, Anderson | $100-$200 |

## TOURER SHOPPING LIST $3,000 - $5,000

| What | Buy This | Cost |
|---|---|---|
| Battery | 200-300Ah Li (Enerdrive/Victron) | $900-$1,800 |
| Solar (roof) | 2x 200W monocrystalline | $400-$700 |
| Solar (portable) | 1x 160-200W folding | $300-$500 |
| Controller | 30-40A MPPT (Victron) | $200-$400 |
| DC-DC charger | REDARC BCDC1240 (40A) | $400-$600 |
| Inverter | 2000W PSW (Enerdrive/Victron) | $350-$600 |
| Monitor | Victron SmartShunt | $130-$160 |
| Wiring + bits | Cable, fuses, Anderson, ext socket | $150-$300 |

## FULL-TIMER SHOPPING LIST $5,000 - $8,000+

| What | Buy This | Cost |
|---|---|---|
| Battery | 300-400Ah Li (Victron/Enerdrive) | $1,800-$3,000 |
| Solar (roof) | 400-600W (2-3 panels) | $600-$1,200 |
| Solar (portable) | 1x 200W+ folding | $400-$700 |
| Controller | 40-50A MPPT (Victron) | $300-$500 |
| DC-DC charger | REDARC BCDC1250D (50A) | $500-$800 |
| Inverter | 3000W PSW (Victron/Enerdrive) | $800-$1,500 |
| Monitor | Victron SmartShunt/BMV-712 | $130-$300 |
| Wiring + bits | Heavy cable, fuses, dist. board | $200-$400 |

SECTION 5

# Checking Your Sizing

BOTTOM LINE

SKIP TO THIS

Buying from the shopping lists in Section 4? Sizing is already done. This is a double-check, or for setups between tiers.

## The Quick Sanity Check

Run these three checks. If all pass, you're good to go.

CHECK 1

### Solar vs Battery

Solar watts should be 1.5-2x your battery Ah. E.g. 200Ah battery = 300-400W solar.

CHECK 2

### Controller vs Solar

Controller amps should be about 1/3 of solar watts. E.g. 400W solar = 30-40A controller.

CHECK 3

### Inverter vs Biggest Appliance

Inverter should be 2x your biggest appliance. E.g. 1000W microwave = 2000W inverter.

**All three checks pass? You're done. Go buy it.**

# Worked Example: Dave and Sarah

Daily use: 2,700 Wh (with 25% buffer). Tourers who drive regularly.

Battery: 2,700 Wh / 12V = 225Ah minimum. With cloudy-day buffer: 400Ah total (2x 200Ah lithium).

Solar: 2,700 / 5.5h / 0.65 = ~760W needed, but DC-DC helps on driving days. 400W roof + 200W portable (600W total).

Controller: 400W / 12V x 1.25 = 42A. They buy a 50A MPPT.

DC-DC: 40A charger gives ~160Ah on a 4-hour drive.

Inverter: Biggest appliance is 1000W microwave. 2000W pure sine wave.

## Dave and Sarah's Final System

| Component | What They Bought |
|---|---|
| Battery | 2x 200Ah lithium (400Ah total) |
| Solar (roof) | 2x 200W panels (400W) |
| Solar (portable) | 1x 200W folding panel |
| Charge controller | 50A MPPT |
| DC-DC charger | 40A |
| Inverter | 2000W pure sine wave |
| Battery monitor | Victron SmartShunt |
| Total cost | ~$4,500 - $5,500 |

## Sun Hours by Region

If you're doing the longer sizing calculation, here are average daily peak sun hours across Australia:

| Region | Good | Poor | Avg |
|---|---|---|---|
| Northern AU (Darwin, Cairns) | 5.5-6.5 | 5.0-6.0 | 5.5-6.0 |
| Central AU (Alice Springs) | 6.5-7.5 | 5.0-5.5 | 6.0-6.5 |
| Eastern (Sydney, Melbourne) | 5.5-6.5 | 3.0-3.5 | 4.5-5.0 |
| Western AU (Perth, Pilbara) | 6.0-7.0 | 3.5-4.0 | 5.0-5.5 |
| South AU (Adelaide, Flinders) | 6.0-7.0 | 3.5-4.0 | 5.0-5.5 |
| Tasmania | 5.0-5.5 | 2.5-3.0 | 3.5-4.0 |

If you travel up north in winter (like most caravanners do), you get excellent sun year-round. Staying in the southeast through winter means less output. Worth sizing up slightly if that's the plan.

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Use the System Sizing Worksheet at the back to run through these calculations with your own numbers.

# Expensive Mistakes

If you've followed this guide, you've already avoided all of these. Read through anyway.

### Mistake 1: Buying AGM Now, Lithium Later

**$400-$800 wasted**

You buy AGM because it's cheaper upfront. Within a year you upgrade to lithium. The AGMs go in the bin. Buy lithium from the start.

### Mistake 2: Not Enough Solar for Your Battery

**$300-$1,000+**

Big battery, not enough solar. It sits undercharged day after day, wearing out years early. Solar should be 1.5-2x your battery Ah.

### Mistake 3: Trusting the Number on the Box

**$500-$1,500**

Your "400W" panels produce 200-280W in real conditions. People who don't know this buy too little and wonder why the battery never fills.

### Mistake 4: Thin Cables

**$200-$500 to rewire**

12V systems need thicker cables than house wiring. Thin cables waste power. Going DIY? One size thicker than you think you need.

### Mistake 5: No Battery Monitor

$500-$1,500

Without a monitor, you guess how much charge is left. You run it down too far, it wears out faster. A SmartShunt costs $130-$160.

### Mistake 6: Modified Sine Wave Inverter

$200-$500

You save $50-$100 on a cheaper inverter that produces rough power. Laptops and chargers stop working properly. Always buy pure sine wave.

### Mistake 7: No External Anderson Plug

$300-$500 wasted

You buy a portable panel but there's nowhere to plug it in outside. An external Anderson socket costs $20-$40 and takes 30 minutes to install.

## The Pattern

Every mistake comes from one of three places:

- **Buying cheap and replacing later. The right gear once is cheaper than the wrong gear twice.**
- **Not knowing real-world performance. Lab specs differ from reality. This guide accounts for that.**
- **Skipping the small stuff. Cables, monitors, and Anderson plugs are cheap. The problems they prevent aren't.**

**If you bought from the shopping lists in Section 4, you've avoided all seven.**

CHECKLIST

# Before You Buy

## THE BASICS

- [ ] Completed the Power Audit (Section 2 or worksheet)
- [ ] Know your daily watt-hours number
- [ ] Identified your tier: Weekender / Tourer / Full-timer

## YOUR GEAR

- [ ] Shopping List worksheet filled in with specific products
- [ ] Battery is lithium with built-in BMS
- [ ] Inverter is pure sine wave
- [ ] Battery monitor included (e.g. Victron SmartShunt)
- [ ] External Anderson plug socket included
- [ ] Cables, fuses, and connectors in your budget

## WILL IT FIT?

- [ ] Measured roof space for panels (clear of A/C, antenna, hatches)
- [ ] Checked battery box dimensions
- [ ] Confirmed weight (lithium is usually lighter than AGM)

## QUICK SIZING CHECK

- [ ] Solar watts = 1.5-2x battery Ah?
- [ ] Controller amps = ~1/3 of solar watts?
- [ ] Inverter handles biggest appliance with room to spare?

## INSTALLATION

- [ ] Got an installer? Hand them the Installer Brief.
- [ ] Going DIY? Grab manufacturer install guides for each component.

CHECKLIST

# What to Tell Your Installer

**Print this page and hand it to your installer.**

## My System Specs

| Component | What I Want |
|---|---|
| Battery | _______ Ah lithium (Brand: _______) |
| Solar (roof) | _______ W total (___ x ___ W panels) |
| Portable panel | _______ W (with ext. Anderson socket) |
| Charge controller | _______ A MPPT (Brand: _______) |
| DC-DC charger | _______ A (Brand: _______) |
| Inverter | _______ W pure sine wave (Brand: _______) |
| Battery monitor | Brand: _______ |

PLEASE MAKE SURE

- [ ] External Anderson plug socket is installed for portable panel
- [ ] Battery monitor is set up and working before you leave
- [ ] All wiring is correctly sized for the system
- [ ] I get a walk-through of the system when done (see next page)

## Questions to Ask

1. "Have you installed this brand/setup before?"
2. "Is there anything in my spec list you'd change, and why?"
3. "What's included in the quote and what's extra?"
4. "How long will it take?" (standard tourer = ~1 day)
5. "What warranty do I get on the installation work?"

## After Installation Walk-Through

Before the installer leaves, get them to show you:

- [ ] How to read the battery monitor (percentage, charging)
- [ ] Where the main fuse/isolator is
- [ ] How to connect the portable panel via Anderson plug
- [ ] System is working: solar charging, battery level going up
- [ ] Test a 240V appliance through the inverter
- [ ] Their contact number for roadside issues

**WHY THIS PAGE EXISTS** SKIP TO THIS

Installers are busy. Walk in with just "I want solar" and you'll get whatever they normally install. This page makes sure you get what you planned for.

CHECKLIST

# Your First Week With Solar

Your system is in. Here's how to get familiar with it. After a week it'll be second nature.

## Day 1: Get to Know Your Monitor

- [ ] Find the battery monitor. Should show a percentage.
- [ ] Turn on your usual stuff (fridge, lights). Watch the monitor.
- [ ] Test the inverter. Phone charger first, then biggest appliance.
- [ ] Plug in portable panel via Anderson socket. Confirm charging.

## Day 2-3: Learn the Pattern

- [ ] Check monitor in the morning. 100% to 85% overnight is normal.
- [ ] Check late afternoon. Back near 95-100%? System is keeping up.
- [ ] Notice the rhythm: drops overnight, fills during the day.

**THE DAILY RHYTHM** SKIP TO THIS

Battery drops overnight (fridge), solar fills it during the day. If the top-up roughly matches drawdown, you're golden. Hovering around 80% by afternoon? Set up the portable panel.

## What You'll See (All Normal)

- Battery sits between 70-100% on a typical day
- Solar output is high midday, low morning/evening, zero at night
- Fridge draws power in bursts (compressor cycling on/off)
- Inverter hums quietly when switched on
- Cloudy days produce less solar (1-2 cloudy days is fine)

## If Something Seems Off

**Battery dropping even though it's sunny?**
Check for shade on panels. Move the van or set up portable panel.

**Inverter switches off when you plug something in?**
Too much at once. Use one big appliance at a time.

**Battery low after a few overcast days?**
Drive for a couple of hours. The DC-DC charger will handle it.

## After a Week

You'll barely think about it. A quick glance at the monitor becomes like checking the fuel gauge. If it stays above 60-70% most of the time, your system is well sized.

**That's it. You've got solar on your caravan and you know how to use it.**

WORKSHEET

# Power Audit

List every electrical appliance you plan to run off-grid. Check the label for wattage. If it shows amps, multiply by 12 to get watts.

| Appliance | Watts | Hrs/Day | Daily Wh |
|---|---|---|---|
| Compressor fridge | 50-70 | 24 | ____________ |
| LED lights (all) | 20-40 | ____ | ____________ |
| TV (12V) | 30-60 | ____ | ____________ |
| Phone charger x ___ | 10-20 ea | ____ | ____________ |
| Laptop charger | 45-65 | ____ | ____________ |
| Tablet charger | 10-15 | ____ | ____________ |
| Water pump | 50-70 | ____ | ____________ |
| Diesel heater fan | 20-40 | ____ | ____________ |
| Range hood fan | 20-30 | ____ | ____________ |
| Radio/speakers | 10-20 | ____ | ____________ |
| CPAP machine | 30-60 | ____ | ____________ |
| Electric blanket | 40-60 | ____ | ____________ |

| Appliance | Watts | Hrs/Day | Daily Wh |
| --- | --- | --- | --- |
| Hair dryer (inverter) | 1200-2000 | ____ | _____________ |
| Microwave (inverter) | 800-1200 | ____ | _____________ |
| Coffee machine (inv.) | 800-1500 | ____ | _____________ |
| Toaster (inverter) | 800-1200 | ____ | _____________ |
| _______________ | _____ | ____ | _____________ |
| _______________ | _____ | ____ | _____________ |
| _______________ | _____ | ____ | _____________ |

**Subtotal:** _______________ Wh

**+ 25% buffer (x 1.25):** _______________ **Wh**

## Your Traveller Tier

**Under 1,500 Wh**
**Weekender**

**1,500-3,000 Wh**
**Tourer**

**Over 3,000 Wh**
**Full-timer**

**My tier:** ________________________

**My daily total:** ______________ Wh

WORKSHEET

# System Sizing

**Daily power use (from Power Audit):** _____________ Wh

STEP 1

## Battery Size

Daily Wh / 12V / 0.85 (lithium usable) = minimum Ah

_______ Wh / 12 / 0.85 = _______ Ah (minimum)

_______ Ah x 1.5 = _______ Ah (with 1 day reserve)

**Your battery choice:** _______ Ah lithium

STEP 2

## Solar Panel Wattage

Daily Wh / peak sun hours / 0.65 (real-world efficiency) = total solar watts

_______ Wh / _______ hours / 0.65 = _______ W

**Fixed panels:** _______ W **Portable:** _______ W

STEP 3

## Charge Controller Size

Fixed panel wattage / 12V x 1.25 (safety margin) = controller amps

_______ W / 12 x 1.25 = _______ A

**Your charge controller:** _______ A MPPT

STEP 4

## Inverter Size

Add up 240V appliances you'd run simultaneously, multiply by 1.2 for surge.

Total simultaneous: _______ W x 1.2 = _______ W

**Your inverter:** _______ W pure sine wave

STEP 5

## DC-DC Charger

Weekender: 20A. Tourer: 40A. Full-timer: 40-60A.

**Your DC-DC charger:** _______ A

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## Complete System Summary

| Component | Spec | Brand/Model | Est. Cost |
|---|---|---|---|
| Battery | ___Ah Li | ____________ | $_______ |
| Solar (fixed) | ___W | ____________ | $_______ |
| Solar (portable) | ___W | ____________ | $_______ |
| Controller | ___A MPPT | ____________ | $_______ |
| DC-DC charger | ___A | ____________ | $_______ |
| Inverter | ___W PSW | ____________ | $_______ |
| Monitor | ________ | ____________ | $_______ |
| Wiring + fuses | ________ | ____________ | $_______ |
| TOTAL | | | $_______ |

QUICK CROSS-CHECK

- ☐ Solar watts = 1.5-2x battery Ah?
- ☐ Controller amps = ~1/3 of solar watts?
- ☐ Inverter = 2-3x biggest 240V appliance?
- ☐ All three pass? System is balanced. Go buy it.

WORKSHEET

# Shopping List

**My tier:** _____________ **Budget range:** $_____________

**Battery**
_______ Ah lithium (LiFePO4)
Must have built-in BMS. Check dimensions fit battery box.
Brand:
Price: $

**Solar (Fixed)**
_______ W total (___ panels x ___ W)
Monocrystalline. Check physical size fits roof.
Brand:
Price: $

**Solar (Portable)**
_______ W folding panel
Check it comes with Anderson plug connector.
Brand:
Price: $

**Charge Controller**
_______ A MPPT
Check max input voltage matches panel config.
Brand:
Price: $

**DC-DC Charger**
_______ A
Check compatibility with vehicle alternator.
Brand:
Price: $

**Inverter**
_______ W pure sine wave
Must be PSW. Check outlet types (GPO, USB).
Brand:
Price: $

**Battery Monitor**
Shunt-based with Bluetooth
Brand:
Price: $

## Wiring and Accessories

| Item | Spec | Qty | Price |
|---|---|---|---|
| Battery cable (pos) | ___ $mm^2$ | ___ m | $_____ |
| Battery cable (neg) | ___ $mm^2$ | ___ m | $_____ |
| Solar cable | $4mm^2$ twin | ___ m | $_____ |
| Anderson plugs (50A) | Standard | ___ pairs | $_____ |
| Ext. Anderson socket | Weatherproof | 1 | $_____ |
| Bus bars (+/-) | Rated for system | 2 | $_____ |
| ANL fuse + holder | Rated for battery | 1 | $_____ |
| Blade fuses (assorted) | Fuse box or individual | ___ | $_____ |
| MC4 connectors | For solar panels | ___ pairs | $_____ |
| Cable ties + clips | Assorted | 1 pack | $_____ |
| Mounting brackets | For panels | ___ sets | $_____ |

## Total Budget

| Category | Estimated Cost |
|---|---|
| Battery | $_____________ |
| Solar (fixed + portable) | $_____________ |
| Charge controller | $_____________ |
| DC-DC charger | $_____________ |
| Inverter | $_____________ |
| Battery monitor | $_____________ |
| Wiring + accessories | $_____________ |
| Total components | $_____________ |
| Installation (if applicable) | $_____________ |
| GRAND TOTAL | $_____________ |

**Use the Before You Buy checklist to make sure you haven't missed anything.**

# You're all set.

You've got your tier, your shopping list, and your checklists. Everything you need to get solar on your caravan without second-guessing yourself.

## YOUR RESOURCES

- **Power Audit Worksheet**
  Work out your daily watt-hours
- **System Sizing Worksheet**
  Size every component to your usage
- **Shopping List Worksheet**
  Track brands, specs, and costs
- **Before You Buy Checklist**
  Final check before you spend
- **Installer Brief**
  Hand this to your installer
- **First Week Checklist**
  Get comfortable with your system

## WHAT TO DO NEXT

1. Complete your Power Audit worksheet
2. Identify your tier (Weekender, Tourer, Full-timer)
3. Fill in your Shopping List from Section 4
4. Run through the Before You Buy checklist
5. Hand your installer the Installer Brief, or order online